

Muhammad Nur Fattah

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Summary

Computer Engineering graduate with a strong interest in machine learning, IoT, embedded systems, and data science. Possesses a solid technical foundation, analytical thinking, and a practical approach to problem solving. A fast learner who adapts quickly to new technologies and methodologies, with a focus on developing effective and efficient technical solutions.

Education

Universitas Brawijaya – Indonesia, Malang **Aug 2022 - Jan 2026**
Computer Engineering, 3.94/4.00
Best Graduate Award – (2026)

Skill

Languages	: Indonesia (Native), English (Basic Conversational)
Programming	: C++, Python, Go, JavaScript, C#
Tools & Tech	: Git, Machine Learning, Computer Vision, IoT, Microcontroller, Backend, Signal Processing, MQTT, Linux, Grafana
Soft Skills	: Fast Learner, Problem Solving, Analytical Thinking, Communication, Collaboration

Work Experience

PT. LINTAS DATA PRIMA Engineering on Site Intern	Madiun, Indonesia Jul 2025 – Sep 2025
- Performed on-site technical support, including network and system implementation across corporate, government, and educational institutions. - Installed, configured, and maintained wireless networks and server infrastructure, including monitoring, troubleshooting, and server relocation activities.	

ROBOTICS & EMBEDDED SYSTEMS LAB Research Assistant Patient Vital Sign Monitor	Malang, Indonesia Sep 2025 – Jan 2026
- Conducted research and development of heart rate detection algorithms as part of a patient vital sign monitoring system. - Implemented a graphical user interface (GUI) for the overall system and conducted system testing in collaboration with medical professionals.	

Research Assistant Auto Drone Development	May 2024 – Apr 2025
- Collected, processed, and organized UAV flight data using Mission Planner for research analysis. - Conducted flight operations and data acquisition to support autonomous drone development.	

Practicum Assistant	Sep 2022 – Nov 2025
- Taught 100+ students in totals for Digital Systems, Microcontroller, and Embedded System practical sessions. - Assisted in teaching Digital Systems (logic gates, flip-flops, multiplexers, registers) - Assisted in Microprocessors & Microcontrollers (I/O, interrupts, timers, communication) - Assisted in Embedded Systems (FSM, RTOS, low-power systems, fault tolerance)	

FILKOM LEARNING LABORATORY
Practicum Assistant

Malang, Indonesia
Aug 2023 – July 2024

- Taught 70+ students in totals for Basic Programming and Data Structure & Algorithm.
- Assisted in teaching basic programming with C++, covering key concepts and coding techniques.
- Assisted in teaching algorithms and data structures, including queues, stacks, trees, etc.

Organizational Experience

ROBOTICS & EMBEDDED SYSTEMS LAB
Student Employee

Malang, Indonesia
Jul 2023 - Dec 2025

- Maintained laboratory operations, equipment management, and practicum support to ensure an effective learning environment.
- Contributed to research, community service projects, and public workshops related to embedded systems and IoT.

Staff of Creative Media

Jul 2023 - Dec 2025

- Designed Instagram content and visual assets using Figma to support branding and information.
- Collaborated with team members to plan and execute digital content strategies to improve audience reach.

Workshop Committee

June 2024 - June 2025

- Teaching Assistant – Workshop “Smart Monitoring with NB-IoT: From MQTT to Grafana Visualization” (2025). Guided in system setup, MQTT communication, and data visualization.
- Teaching Assistant – Workshop on “Embedded Systems and Cyber-Physical Systems” (2024).

Award and Certification

Award

- Best Computer Engineering Graduate **2026**
- Finalist – Sustainability Sprint Hackathon, Penang, Malaysia **2024**

Certification

- EF SET English Certificate 54/100 B2 Upper Intermediate **2026**
- Big Data 101, Hadoop 101 – by IBM Learning **2026**
- Belajar Machine Learning untuk Pemula -- by Dicoding Indonesia **2026**
- Dasar Data Science -- by Dicoding Indonesia **2026**
- Fundamental Analisis Data – by Dicoding Indonesia **2026**
- Belajar Dasar Structured Query Language (SQL) -- by Dicoding Indonesia **2026**
- Microsoft Office Desktop Application – by Trust Training Partners **2025**

Project

Arrhythmia Detection

2026

- Developed an ECG signal processing system using Pan-Tompkins++ algorithm for real-time heart rate (BPM) calculation for arrhythmia detection, including signal visualization and hardware integration.

Parking Management System

2026

- Developed a full-stack automated parking system using Golang, React, PostgreSQL, Redis, Docker, and YOLO-based vehicle & license plate detection for automatic vehicle identification, entry recording, and slot monitoring.

Market Trend Analysis

2026

- Conducted market trend analysis through data cleaning, exploratory data analysis (EDA), and visualization. Developed an interactive Streamlit dashboard to present market trends and analytical insights.

Fatigue Detection System

2025

- Developed a wearable fatigue automated detection system, implementing embedded systems and machine learning models on ESP32 with MQTT-based data transmission.

Face Detection

2025

- Developed a custom face detection model using Roboflow for dataset labeling, segmentation, and training, capable of accurately detecting personal facial features in real-time.

Automation Server Project – Ansible & Terraform

2024

- Developed an automation system for high-availability server management, including Nginx configuration, database replication, load balancing, and DNS automation.

Apps Development – Ecoplants

2023

- Participated in a workshop where I developed a plant e-commerce application within one month, serving as the backend developer using Go language.